

Class 10 Science - Chapter 2

Acids, Bases and Salts (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)

Class 7-8-9 ki yeh cheezein pehle pakki karo, warna Ch2 samajh nahi aayega.
(Yeh pura RED page hai - revision ke time sabse pehle ise dekho.)

1. Acid aur Base - Class 7 wali pehchaan

Acid = khatte (sour) swaad wale padarth. Jaise: nimbu, imli, sirka (vinegar), dahi. "Acid" Latin shabd 'acidus' = khatta se aaya hai.

Base = kadwe (bitter) swaad wale aur chhoone par CHIKNE (soapy) lagte hain.
Jaise: baking soda, sabun, chuna paani.

Indicator = woh padarth jo acid/base me rang badal kar batata hai.

Natural indicator: Litmus (lichen se banta), Haldi, China rose (gudhal).

2. Litmus ka rule (sabse base concept)

Litmus do colour me aata: BLUE litmus aur RED litmus.

Acid -> BLUE litmus ko RED kar deta hai.

Base -> RED litmus ko BLUE kar deta hai.

Yaad rakho: "Acid me Blue Roye (B->R), Base me Red Blue ho jaaye (R->B)".

Neutral (jaise distilled water) kisi litmus ka rang nahi badalta.

3. Atom, Ion, Molecule (Class 8-9 base)

Atom = sabse chhota kann. Ion = charge wala atom/group.

Cation = +ve charge (electron khoya). Anion = -ve charge (electron mila).

H⁺ (hydrogen ion), OH⁻ (hydroxide ion), Cl⁻, Na⁺, SO₄²⁻, CO₃²⁻.

Molecule = atoms ka group jud kar bana. HCl, H₂O, NaOH.

Compound = do ya zyada alag elements chemically jude.

4. Common Formula yaad rakho (ratta zone)

Acids: HCl (hydrochloric), H₂SO₄ (sulphuric), HNO₃ (nitric),
H₂CO₃ (carbonic), CH₃COOH (acetic - sirke me).

Bases: NaOH (sodium hydroxide), KOH (potassium hydroxide),
Ca(OH)₂ (calcium hydroxide - chuna), NH₄OH (ammonium hydroxide).

Salts: NaCl (common salt), Na₂CO₃ (washing soda), NaHCO₃ (baking soda),
CaCO₃ (limestone/marble), CuSO₄ (copper sulphate).

5. Reaction ki bhasha (Class 9 revision)

Reactant (arrow ke baayi taraf) ----> Product (arrow ke daayi taraf).

States: (s)=solid, (l)=liquid, (g)=gas, (aq)=aqueous (paani me ghula).

Equation hamesha BALANCE honi chahiye (dono taraf atom barabar).

Effervescence = bubbles/jhaag nikalna (gas banne ki nishani).

6. Concentrated vs Dilute aur ek SAFETY rule

Concentrated = paani kam, acid/base zyada (strong/khatarnak).

Dilute = paani zyada daal kar patla kiya hua.

SAFETY (bahut important): "Acid ko paani me daalo, paani ko acid me NAHI".

Kyunki paani+acid milte waqt bahut garmi nikalti, acid me paani daalo to
chhinta ud kar jala sakta hai.

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT

Class 10 Science | Ch 2: Acids, Bases and Salts

[GREEN = board exam me baar-baar poocha jaata hai]

1. Indicators (Suchak) - poora list

Indicator acid/base ki maujoodgi batata hai (rang/smell badal kar).

Types:

- (1) Natural: Litmus (purple/blue/red), Haldi (turmeric), China rose.
- (2) Synthetic: Methyl orange, Phenolphthalein.
- (3) Olfactory indicator: SMELL se batata (jaise pyaaz, vanilla, laung).
Base me inki smell chali jaati, acid me bani rehti.

Indicator colour table (RATTA - exam me direct aata):

Indicator	Acid me	Base me
-----	-----	-----
Blue litmus	Red ho jaata	Blue hi rehta
Red litmus	Red hi rehta	Blue ho jaata
Methyl orange	Red / Pink	Yellow
Phenolphthalein	Colourless	Pink
Haldi (turmeric)	Yellow hi	Reddish-brown

2. Acids/Bases ke Chemical Properties [TOP EXAM AREA]

(A) Metal ke saath reaction

Acid + Metal ----> Salt + Hydrogen gas (H₂)



Test of H₂ gas: jalti hui teeli (candle) paas le jaao to "POP" sound aati.

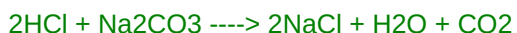
Base + Metal (kuch active metals) ----> Salt + Hydrogen gas



(B) Metal carbonate / bicarbonate ke saath

Acid + Metal Carbonate ----> Salt + CO₂ + H₂O

Acid + Metal Bicarbonate ----> Salt + CO₂ + H₂O



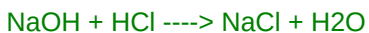
CO₂ test: CO₂ ko chuna paani (lime water) me se nikalo to woh DOODHIYA

(milky) ho jaata. $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 \text{ (white)} + \text{H}_2\text{O}$.

Zyada CO₂ daalo to milkiness gayab ($\text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{Ca(HCO}_3)_2$).

(C) Acid + Base = NEUTRALISATION [VERY IMPORTANT]

Acid + Base ----> Salt + Water (yeh NEUTRALISATION reaction hai)



Iska asli khel: H^+ (acid se) + OH^- (base se) $\rightarrow \text{H}_2\text{O}$.

(D) Metal oxide aur Non-metal oxide

Acid + Metal oxide $\rightarrow$ Salt + Water (Metal oxides BASIC hote hain)



Base + Non-metal oxide $\rightarrow$ Salt + Water (Non-metal oxides ACIDIC hote hain)



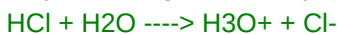
Conclusion: Metal oxide = basic ; Non-metal oxide = acidic.

3. Acid/Base me aisa kya common hai? [VERY IMPORTANT]

Saare acids me H^+ (hydrogen) ion COMMON hota hai.

Saare bases me OH^- (hydroxide) ion COMMON hota hai.

Acid paani me ghulkar H^+ (asal me H_3O^+ , hydronium) deta hai.



IMPORTANT: Acid sirf PAANI ki maujoodgi me hi H^+ deta hai (ionise hota).

Isiliye DRY HCl gas, dry blue litmus ka rang NAHI badalti.

Base jo paani me ghulta hai use ALKALI kehte hain (NaOH, KOH).

4. Acid/Base ko paani me ghol-na (Dilution)

Acid ya base ko paani me ghol-ne par H^+/OH^- ki concentration GHAT-ti hai.

Yeh process EXOTHERMIC hai (garmi nikalti).

SAFETY: "Hamesha acid ko paani me daalo (slowly + stir), ulta KABHI nahi" -
warna achanak bahut garmi se chhinta ud sakta hai (acid splash).

5. Strength: pH Scale [SABSE IMPORTANT TOPIC]

pH = "potenz of Hydrogen" - H^+ ion ki concentration naapne ka scale.

Scale 0 se 14 tak:

pH = 7 $\rightarrow$ Neutral (jaise pure/distilled water)

pH < 7 $\rightarrow$ Acidic (jitna kam, utna strong acid). 0 = strongest acid.

pH > 7 $\rightarrow$ Basic / Alkaline (jitna zyada, utna strong base). 14 = strongest base.

Universal indicator: alag-alag pH par alag rang deta (poora range).

Strong vs Weak:

Strong acid (HCl, H_2SO_4 , HNO_3) $\rightarrow$ paani me PURA ionise (zyada H^+).

Weak acid (CH_3COOH , H_2CO_3) $\rightarrow$ THODA ionise (kam H^+).

Strong base (NaOH, KOH) ; Weak base (NH_4OH , $\text{Mg}(\text{OH})_2$).

pH daily life me (board favourite):

Body/blood pH $\sim$ 7.4 (bahut narrow range me rehna chahiye).

Acid rain pH < 5.6 $\rightarrow$ nadiyon ka pH girta $\rightarrow$ jal-jeevan ko nuksaan.

Mitti (soil) ka pH theek hona zaroori - acidic mitti me chuna daalte.

Pet (stomach) me HCl $\rightarrow$ jyada ho to acidity $\rightarrow$ ANTACID (base jaise

Milk of Magnesia / $\text{Mg}(\text{OH})_2$) lete hain neutralise karne ko.

Daant (teeth): muh ka pH 5.5 se neeche jaaye to enamel ghisne lagta;

bacteria sugar se acid banate $\rightarrow$ isiliye base wala toothpaste use karo.

Bee/cheenti ka doonk -> methanoic (formic) acid -> baking soda (base) lagao.

Nettle plant doonk -> formic acid -> dock plant patta (base) ragdo.

6. SALTS [BADA AUR SCORING TOPIC]

Salt = acid aur base ke neutralisation se banta padarth.

pH of salts (kis acid+base se bana, us par depend):

Strong acid + Strong base -> NEUTRAL salt (pH = 7). Eg: NaCl.

Strong acid + Weak base -> ACIDIC salt (pH < 7). Eg: NH₄Cl.

Weak acid + Strong base -> BASIC salt (pH > 7). Eg: Na₂CO₃, CH₃COONa.

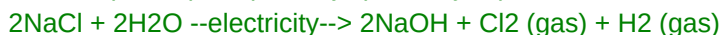
7. Important Salts (har ek board me aata) [VERY IMPORTANT]

(A) Common Salt - NaCl

Samudra ke paani + rock salt se. Yeh kayi cheezon ka "raw material" hai.

(B) Sodium Hydroxide - NaOH (CHLOR-ALKALI process)

NaCl ke paani (brine) me bijli (electrolysis) chalate hain:



3 product 3 jagah use: NaOH (soap), Cl₂ (bleaching/PVC), H₂ (fuel).

Cathode par H₂, Anode par Cl₂, solution me NaOH banta.

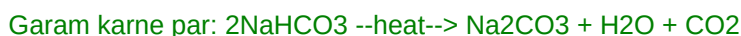
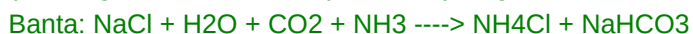
(C) Bleaching Powder - CaOCl₂ (Calcium oxychloride)

Slaked lime par Cl₂ gas paas karke banta:



Use: kapde/paani bleach + disinfect, peene ke paani ko saaf karna.

(D) Baking Soda - NaHCO₃ (Sodium hydrogen carbonate / bicarbonate)



Use: baking (cake fulaata - CO₂ nikalta), antacid (pet ki acidity),
soda-acid fire extinguisher, baking powder ka hissa.

Baking powder = baking soda + ek mild edible acid (jaise tartaric acid).

Tartaric acid CO₂ ke saath bani Na₂CO₃ ko neutralise karta (kadwa nahi lagta).

(E) Washing Soda - Na₂CO₃·10H₂O (sodium carbonate decahydrate)



Use: ghar ki safai, kaanch/sabun banana, paani ki "permanent hardness" hatana.

8. Water of Crystallisation aur Plaster of Paris [IMPORTANT]

Water of crystallisation = salt ke crystal me FIXED maatra me juda paani.

Eg: CuSO₄·5H₂O (blue vitriol) me 5 water; Na₂CO₃·10H₂O me 10; CaSO₄·2H₂O (gypsum).

Blue CuSO₄·5H₂O ko garam karo -> safed ho jaata (paani udd jaata),

phir paani daalo -> wapas neela (water of crystallisation wapas).

Plaster of Paris (POP) = CaSO₄·(1/2)H₂O (calcium sulphate hemihydrate).

Gypsum (CaSO₄·2H₂O) ko 373K (100 C) tak garam karke banta:



POP + paani -> wapas gypsum ban kar SAKHT (set) ho jaata (hardening).

Use: tooti haddi ka plaster (doctor), murti/khilone, deewar smooth karna.

NOTE: POP ko nami se door (airtight) rakho, warna set ho jaayega.

9. Quick Revision - One Liners

- Acid: H^+ deta, blue litmus red, $pH < 7$.
- Base/Alkali: OH^- deta, red litmus blue, $pH > 7$.
- Neutralisation: Acid + Base $\rightarrow$ Salt + Water.
- Acid + metal $\rightarrow$ salt + H_2 (pop test).
- Acid + carbonate $\rightarrow$ salt + CO_2 + water (lime water milky).

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Step-by-step. Pehle mushkil, taaki baad me sab aasaan lage.

Example 1 (HARDEST) - Chlor-alkali + saare products ka use

Q: NaCl ke jaleeya (aqueous) ghol me bijli chalane par kya banta hai? Equation, electrode par kahan kya, aur teeno products ke use likho.

Solution:

Reaction: $2\text{NaCl} + 2\text{H}_2\text{O} \xrightarrow{\text{electricity}} 2\text{NaOH} + \text{Cl}_2 + \text{H}_2$

- Anode (+) par: Cl_2 gas nikalti.

- Cathode (-) par: H_2 gas nikalti.

- Solution me: NaOH (sodium hydroxide) bachta.

Uses: NaOH -> soap/detergent, paper. Cl_2 -> water treatment, PVC, bleaching powder. H_2 -> fuel, margarine, ammonia banana.

Iska naam Chlor-alkali process ($\text{Chlor} = \text{Cl}_2$, $\text{alkali} = \text{NaOH}$).

Example 2 - Water of crystallisation wala observation

Q: Blue CuSO_4 crystal ko garam karne par kya hota? Wapas paani daalo to?

Solution:

- Garam karne par paani (water of crystallisation) udd jaata -> crystal SAFED.

$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \xrightarrow{\text{heat}} \text{CuSO}_4 (\text{white}) + 5\text{H}_2\text{O}$

- Thande hone par paani ki kuch boondein daalo -> wapas NEELA ho jaata, kyunki water of crystallisation wapas jud jaata.

Seekh: rang neela isi 5 water of crystallisation ki wajah se hai.

Example 3 - POP banana + ek hi cheez do roop

Q: Plaster of Paris kaise banta? Formula? POP set kyun ho jaata hai?

Solution:

- Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) ko $\sim 373\text{K}$ tak garam karo:

$\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{\text{heat}} \text{CaSO}_4 \cdot (1/2)\text{H}_2\text{O} + (3/2)\text{H}_2\text{O}$ [yeh POP hai]

- POP me paani milao to wapas gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) ban jaata aur sakht (hard) ho jaata -> isliye haddi/murti ke liye use hota.

Example 4 - Antacid kaam kaise karta (neutralisation)

Q: Pet me acidity (jyada HCl) hone par antacid kaise rahat deta hai?

Solution:

- Antacid ek MILD base hota (jaise Milk of Magnesia = $\text{Mg}(\text{OH})_2$).

- Base, extra HCl ko neutralise kar deta -> Salt + Water banta -> jalan kam.

$\text{Mg}(\text{OH})_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O}$

- Yeh neutralisation reaction ka daily-life example hai.

Example 5 (EASIEST) - Litmus + ek simple equation

Q: (a) HCl blue litmus ka rang kya karega? (b) Zn + HCl ka product?

Solution:

(a) HCl ek acid hai -> Blue litmus ko RED kar dega.

(b) $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$ (gas). Gas jalane par "pop" sound deti.

PART D - SELF TEST (EASY -> HARD)

Khud likho, phir niche answer hint se check karo. Active recall = best memory.

EASY (1 mark)

- Q1. Acid aur base ko litmus se kaise pehchaante hain?
- Q2. $\text{pH} = 7$, $\text{pH} < 7$, $\text{pH} > 7$ ka matlab batao.
- Q3. Neutralisation reaction ki definition + general equation.
- Q4. Baking soda aur washing soda ke chemical formula likho.

MEDIUM (2-3 marks)

- Q5. Acid ke metal carbonate ke saath reaction ka general + 1 example.
CO₂ ko kaise test karoge?
- Q6. Olfactory indicator kya hai? 2 example.
- Q7. "Acid ko paani me daalo, paani ko acid me nahi" - kyun? Samjhao.
- Q8. Strong acid aur weak acid me 1 difference + 1-1 example.

HARD (3-5 marks, board favourite)

- Q9. Chlor-alkali process: equation, electrode products aur teeno ka use.
- Q10. Plaster of Paris: formula, banane ki vidhi (equation), 2 use, ek savdhani (storage).
- Q11. pH ka daily life me mahatva: (a) dant (b) pet (c) acid rain
(d) cheenti/bee doonk - har ek explain karo.
- Q12. Water of crystallisation kya hai? CuSO₄.5H₂O ko garam karne aur phir paani daalne par kya hota - equation sahih.

ANSWER HINTS (PART D)

- A1. Acid: blue litmus -> red. Base: red litmus -> blue.
- A2. 7 = neutral; <7 = acidic; >7 = basic/alkaline.
- A3. Acid + Base -> Salt + Water (eg NaOH + HCl -> NaCl + H₂O).
- A4. Baking soda = NaHCO₃; Washing soda = Na₂CO₃.10H₂O.
- A5. Acid + carbonate -> salt + CO₂ + H₂O (2HCl+Na₂CO₃->2NaCl+H₂O+CO₂);
CO₂ lime water ko doodhiya kar deta.
- A6. Smell se acid/base batane wala (pyaaz, vanilla, laung).
- A7. Dilution exothermic hai; acid me paani daalo to garami+chhinta -> jal sakte.
- A8. Strong (HCl) pura ionise; Weak (CH₃COOH) thoda ionise.
- A9. 2NaCl+2H₂O--elec-->2NaOH+Cl₂+H₂; anode Cl₂, cathode H₂, soln NaOH;
uses soap/bleach/fuel.
- A10. CaSO₄.(1/2)H₂O; gypsum 373K garam; haddi plaster + murti; nami se door.
- A11. Dant: pH<5.5 enamel ghise; Pet: HCl jyada=acidity, antacid; acid rain
pH<5.6 jal-jeevan nuksaan; doonk formic acid, base(baking soda) lagao.
- A12. Crystal me juda fixed paani; CuSO₄.5H₂O garam->white(+5H₂O), paani

daalo->wapas blue.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

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